Credit Cards & Debt — Cheat Sheet

Financial Algebra · Unit 4: Credit Cards & Debt

Essential vocabulary

APR (annual percentage rate)
The yearly cost of borrowing expressed as a percent, used to compute interest on a balance

Periodic rate
The APR divided into the rate for one billing period or day

Balance
The amount owed on an account at a given time

Grace period
The time between the end of a billing cycle and the due date during which new purchases are not charged interest if the full balance is paid

Average daily balance
A method of computing interest by averaging the balance owed on each day of the billing cycle

Minimum payment
The smallest payment required by the due date to keep the account in good standing

Credit limit
The maximum balance a lender allows on a credit account

Credit utilization
The percent of available revolving credit currently being used

Credit score
A number (commonly 300–850) summarizing credit risk based on payment history, amounts owed, length of history, new credit, and credit mix

Credit report
A record of a person's credit accounts and payment history kept by credit bureaus

Late fee
A charge for paying less than the minimum by the due date

Debt avalanche
A repayment strategy that sends extra money to the debt with the highest interest rate first

Debt snowball

Key formulas

Monthly rate	$\frac{\text{APR}}{12}$
Monthly interest	$\text{balance} \times \frac{\text{APR}}{12}$
Average daily balance	$\frac{\text{sum of (balance} \times \text{days)}}{\text{days in cycle}}$
Payment to finish in n months	$\text{PMT} = \frac{P \cdot i}{1 - (1 + i)^{(-n)}}$
Utilization	$\frac{\text{balances}}{\text{limits}} \times 100$

Repayment strategies

Avalanche	Highest APR first → least interest
Snowball	Smallest balance first → quick wins
Balance transfer	Fee + promo length + later APR
Minimum only	Slowest and most expensive

Worked example

1. \$3,000 at 24.99% APR
2. First month interest = $3{,}000 \times \frac{0.2499}{12} =$ \$62.48
3. Minimum (1% + interest) = \$92.48
4. Fixed \$150/month → about 26 months; minimum-only → many years

A repayment strategy that sends extra money to the smallest balance first for quick wins

Balance transfer
Moving a balance to another card, often at a low promotional rate with a transfer fee

Cash advance
Withdrawing cash using a credit card, usually with a fee, a higher APR, and no grace period

Revolving credit
Credit that can be borrowed, repaid, and borrowed again up to a limit

Common misconceptions

- APR is charged monthly as-is.
- Minimum payment is the recommended payment.
- Carrying a balance builds credit.
- 0% transfer offers are always free.

Questions to ask before deciding

- What is the APR — and what is the monthly rate?
- Can I pay the full balance by the due date?
- What fees apply (annual, late, cash advance, transfer)?
- How long will repayment take at my planned payment?
- How will this affect my utilization?

Decision tips

- Pay the full statement balance whenever possible.
- Set autopay for at least the minimum to avoid late fees.
- Keep utilization low.
- Avoid cash advances.

VALUES AND RATES ARE ILLUSTRATIVE. ALL EXAMPLES ARE FICTIONAL. MR. K'S CLASSROOM · FINANCIAL ALGEBRA